

Ismail Aissaoui
Chlef, Algeria
Phone: +213 660 70 77 96
Email: ismail.aissaoui.pro@gmail.com

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Jérôme David
mOeX Team
Inria Montbonnot
Grenoble, France

Dear Dr. David,

I am writing to express my strong interest in the PhD position on "Multi-Perspective Data Interlinking: Combining Standpoint Logic and Link Keys for the Digital Twin of France" within the mOeX team at Inria. As an AI & Data Science Engineer with a deep specialization in knowledge representation, Digital Twins, and ontology alignment, I am excited by the scientific challenge of reconciling conflicting and evolving semantic conceptualizations within the JUNN infrastructure.

During my work on the "Cardiovascular Medical Digital Twin," I engineered complex multi-scale data fusion architectures capable of handling heterogeneous streams of high-frequency sensor telemetry (100Hz VitalDB). Just as historical and administrative metadata in the JUNN project evolve over time, physiological data requires rigorous tracking of changing "standpoints" or states. I addressed this by developing a custom Missingness-Aware Gate to handle sensor data heterogeneity and missingness dynamically. My robust background in semantic embeddings (utilizing PyTorch Geometric and SBERT for NLP text processing) directly aligns with the ontology matching and semantic reconciliation required to interlink disparate territorial knowledge graphs.

I am particularly drawn to your proposal to use standpoint logic (SL) to allow data providers to maintain their own partial interpretations without forcing a unique consensus. My Master's thesis involved complex mathematical modeling and physics-informed neural networks (PI-DeepONet) to enforce rigorous constraints across differing sub-domains (Branch/Trunk Nets). I am eager to apply this rigorous mathematical and architectural mindset to formalize identity links and standpoint link keys for RDF data interlinking.

Enclosed are my CV and academic transcripts for your review. I would be thrilled to bring my expertise in deep learning, data pipelines, and multi-scale Digital Twins to the mOeX team and contribute to the JUNN initiative. I look forward to the possibility of discussing how my technical background and strong autonomy align with your research vision.

Sincerely,

Ismail Aissaoui